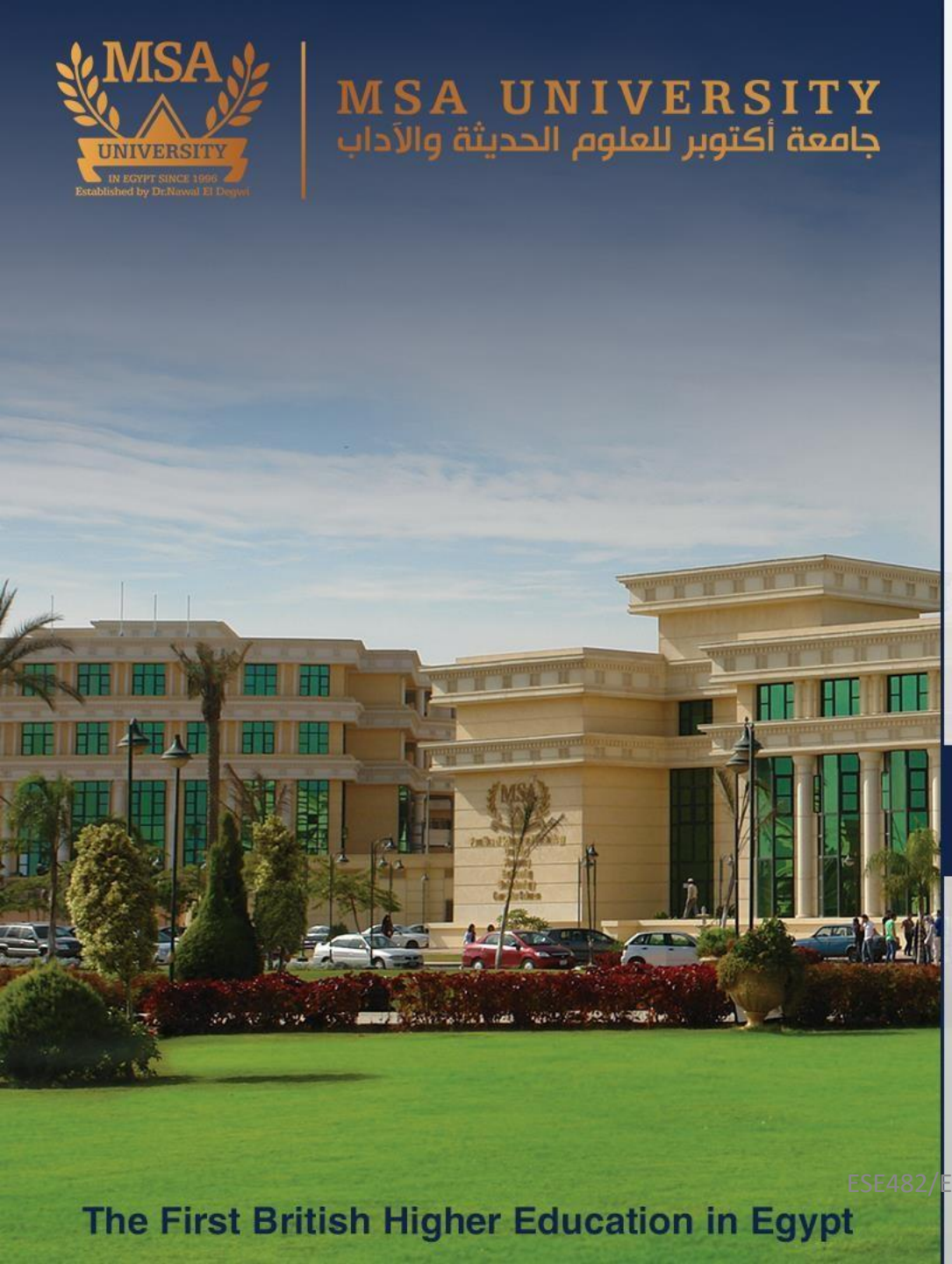


Avionics Navigation Systems

ESE482\ECE4635 - Fall 2026

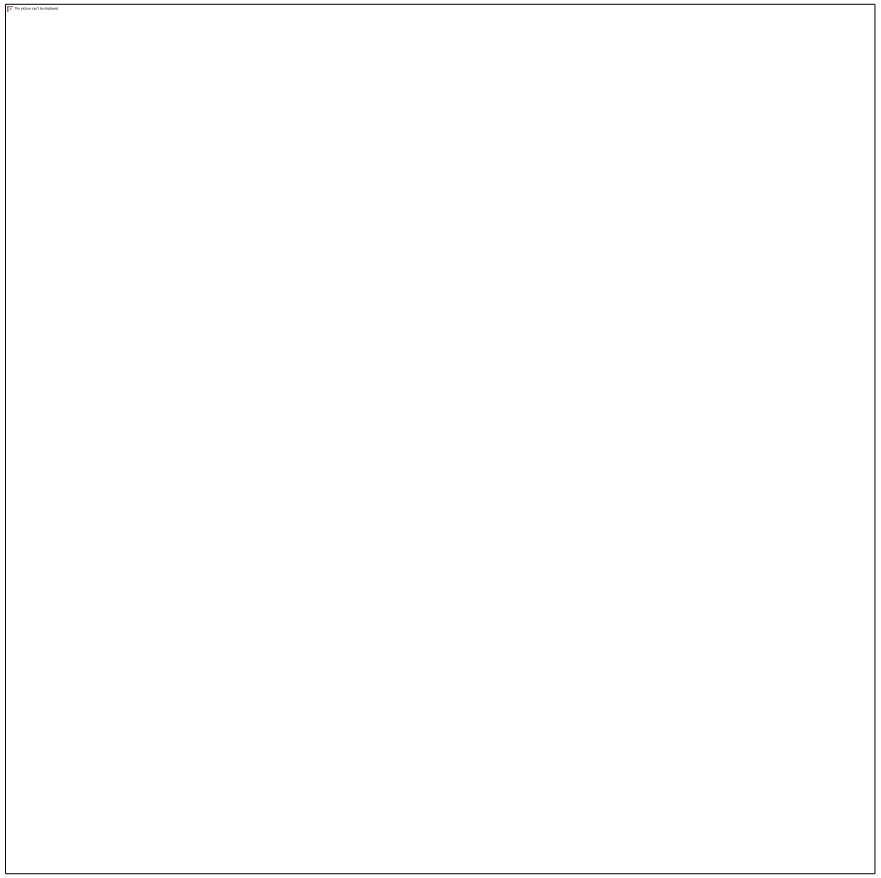
Lecture 5

Dr. Farah Raad

Lecture Topic

Terrestrial Radio-Navigation Systems

Chapter # 4



Introduction

Introduction

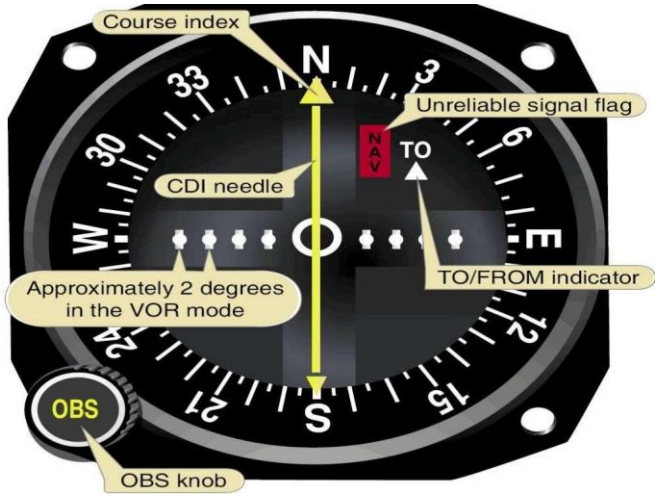
How can an aircraft determine its position, bearing, or distance using radio signals transmitted from terrestrial stations?

What are Terrestrial Radio-Navigation Systems?

Terrestrial Radio-Navigation Systems are navigation systems that help an aircraft determine its position, direction, or distance from a reference point using radio signals transmitted by ground stations at known locations.

- Imagine an aircraft flying from Cairo to London. Instead of relying only on visual observations or the pilot's estimates, the aircraft can receive signals from ground stations and use these signals to calculate its position relative to those stations.







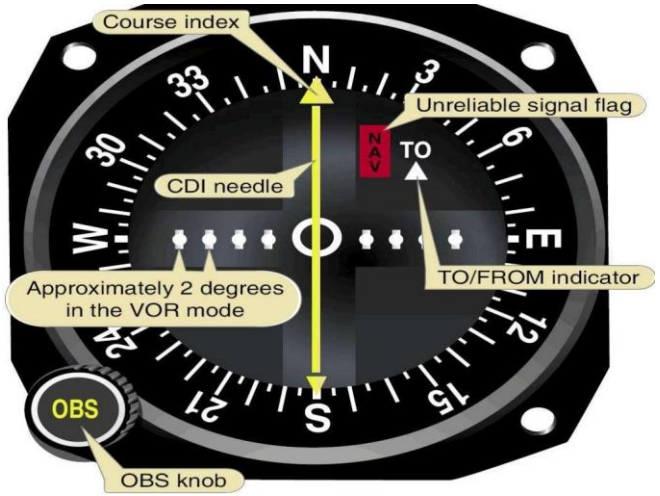
Introduction

- **The basic idea:** The ground station transmits a signal to the aircraft or receives a signal from it.

The navigation equipment installed on the aircraft then uses this information to determine one of the following:

- Bearing: The direction of the aircraft relative to a ground station.
- Distance: The distance between the aircraft and the ground station.
- Position: The location of the aircraft, using information from one or more stations, depending on the system.
- Track Guidance: Guiding the aircraft along a specified flight path.

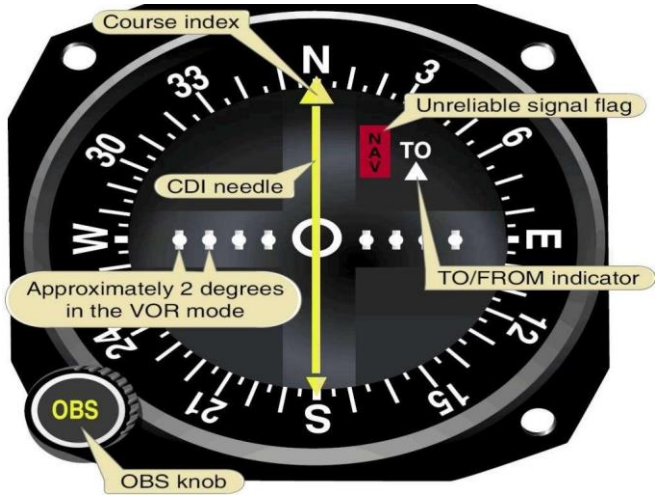






Introduction

- **The navigation process usually consists of three steps:**
- ❑ The ground station transmits a radio signal or responds to an interrogation signal transmitted by the aircraft.
- ❑ The aircraft's equipment measures a specific property of the signal, such as the **phase difference, travel time, or angle of arrival**.
- ❑ The navigation equipment converts the measurement into useful information for the pilot, such as the bearing, distance, or deviation from the desired flight path.
- ✓ Important Note: Not all terrestrial navigation systems operate in the same way. Some rely on bearing measurements, others on distance measurements, and some combine more than one type of information.





Radio Transmission and Reception

- how a radio signal travels from a ground station to an aircraft?

A. Radio Frequency and Wavelength

The fundamental relationship between frequency and wavelength is:

$$v = f\lambda$$

Example:

If the signal frequency is $f = 120\ \text{MHz}$, calculate the wavelength.

Solution:

Using the relationship between wavelength, the speed of light, and frequency:

$$\lambda = \frac{c}{f}$$

Substituting the given values:

$$\lambda = \frac{3 \times 10^8}{120 \times 10^6}$$

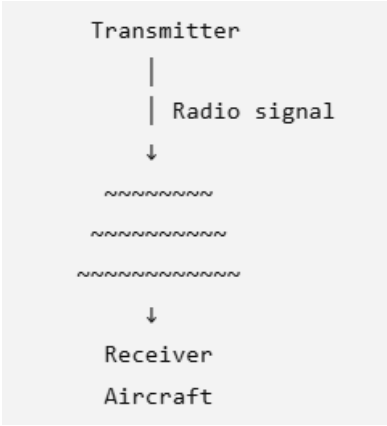
$$\boxed{\lambda = 2.5\ \text{m}}$$

$$v = f\lambda = 120 \times 2.5 = 300\ \text{m/s}$$

Radio Transmission and Reception

➢ **The basic radio-navigation system can be represented as:**

- The transmitter generates an electromagnetic signal.
- The signal propagates through space or along the Earth's surface and eventually reaches the aircraft antenna.
- The receiver then extracts the information contained in the signal.



❑ **Continuous-wave vs pulse transmission**

One important distinction is whether the transmitter sends:

- **Continuous wave**

The signal is transmitted continuously: ————————————————————

- **Pulsed signal**

The transmitter sends short pulses: ____|¯|_____|¯|_____|¯|____

Different navigation systems use different forms.

❖ **For example**, **DME** relies on pulse timing, while systems such as **VOR** use continuous radio signals with modulation.

Modulation:

- amplitude modulation — AM
- frequency modulation — FM
- phase modulation — PM

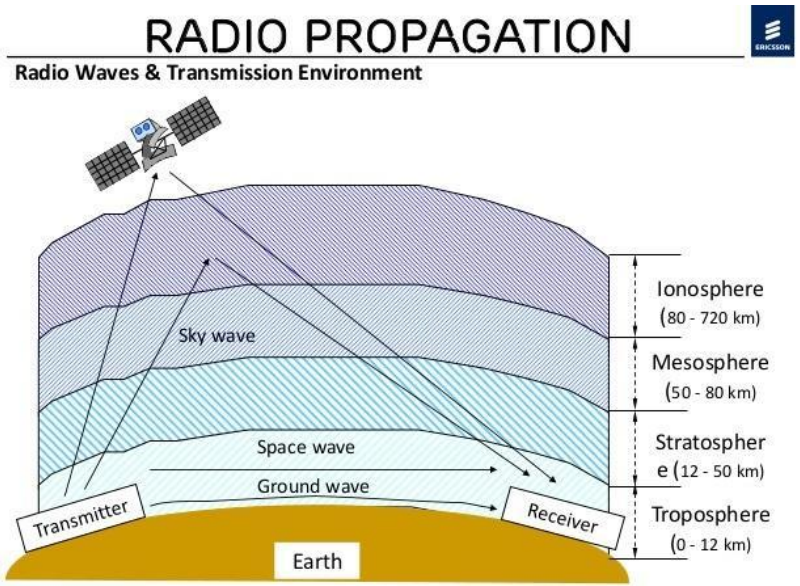
The choice affects how information is encoded and recovered.

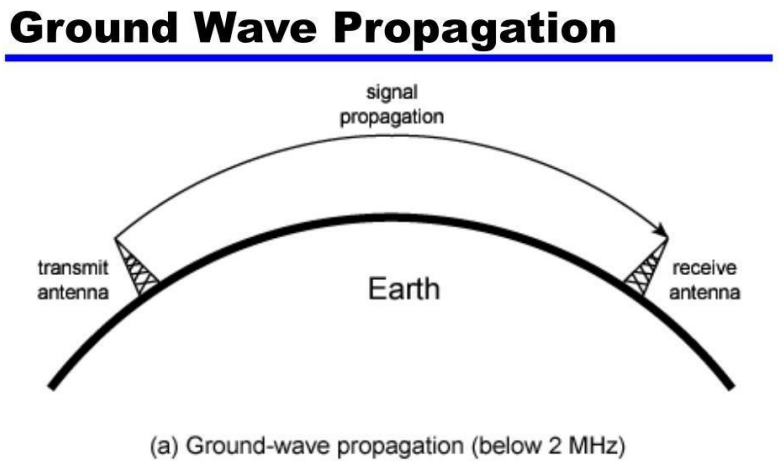
Wave Propagation Methods

Radio waves can travel from a ground station to an aircraft in different ways, depending on **the signal frequency, the environment, and the characteristics of the Earth's atmosphere**.

1. Ground-Wave Propagation

Ground waves travel close to and along the Earth's surface.
They are used in some low-frequency radio systems, and their range depends on the ground's electrical properties and the signal frequency.





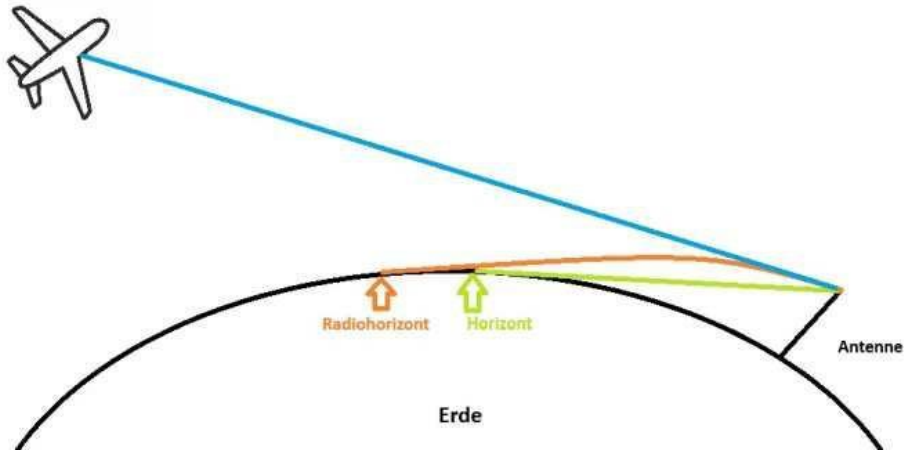
(a) Ground-wave propagation (below 2 MHz)

Wave Propagation Methods

2. Line-of-Sight Propagation

In line-of-sight propagation, the radio signal travels approximately along a direct path between the aircraft and the ground station.
The range of these systems depends on the heights of the aircraft and the ground antenna. Mountains, buildings, and other obstacles can block the signal.

Example: VOR and DME are commonly used aircraft navigation systems that operate primarily with line-of-sight radio propagation.



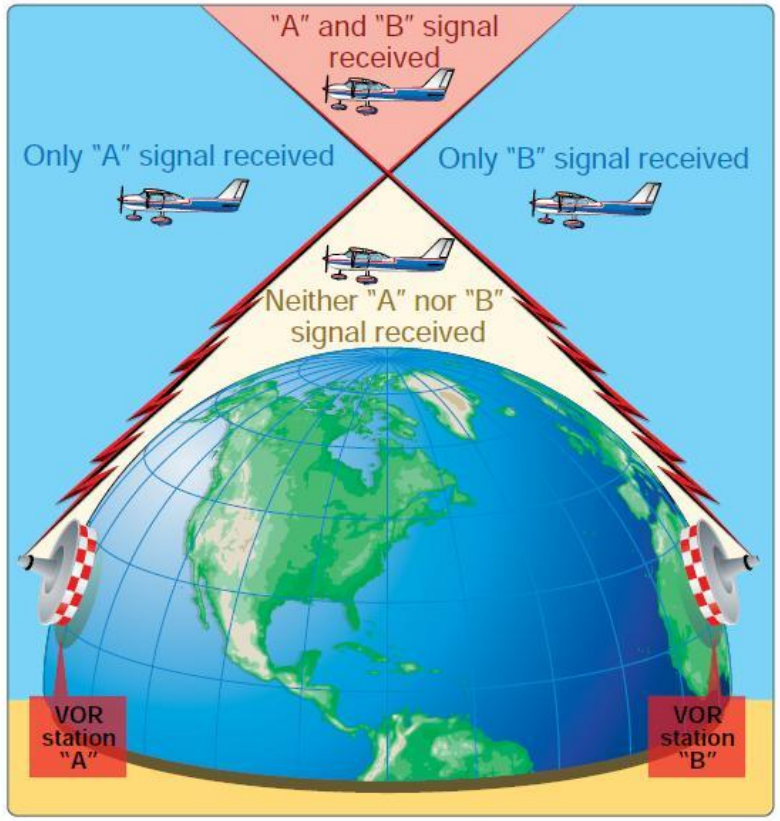


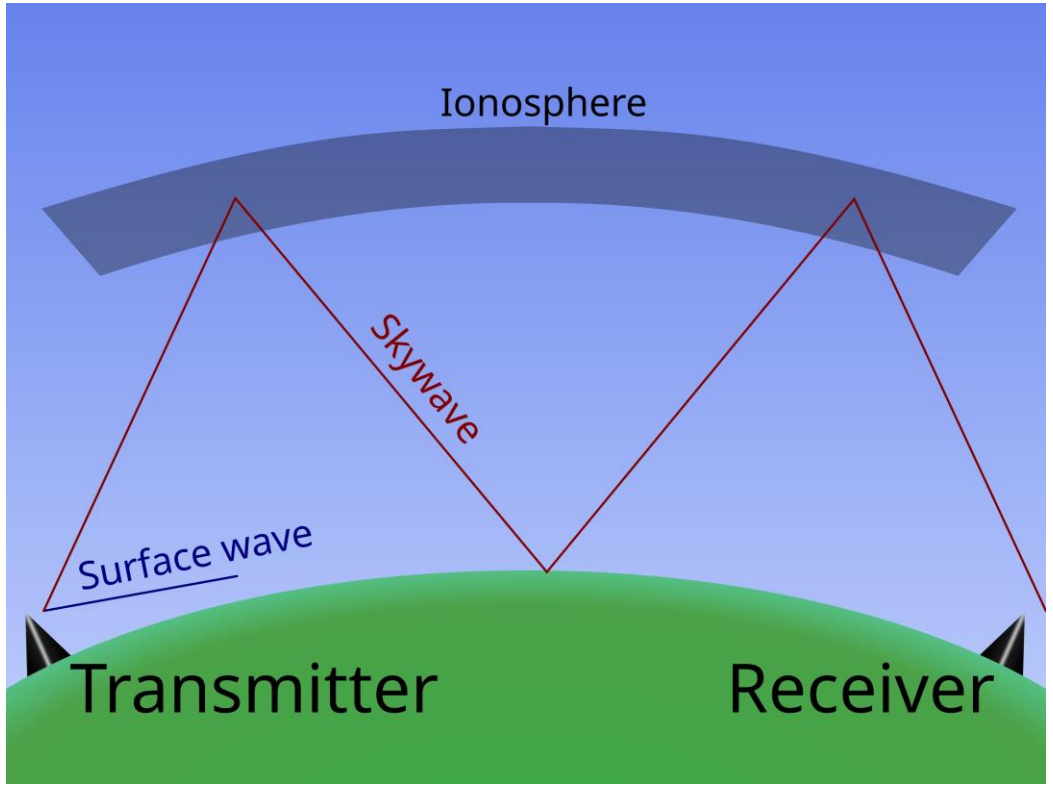
Figure 16-28. *VHF transmissions follow a line-of-sight course.*

Wave Propagation Methods

3. Skywave Propagation

In skywave propagation, certain radio frequencies can be refracted by the ionosphere and return toward the Earth's surface. This allows radio signals to travel beyond the normal line-of-sight range.

This principle is important for understanding long-distance radio communication and some long-range navigation systems.



Main Sources of Noise and Propagation Errors

Several factors can affect the quality of a received radio signal and reduce navigation accuracy.

1. Thermal Noise: Noise caused by the random thermal motion of electrons in electronic circuits and components.
2. Atmospheric Noise: Natural interference caused by atmospheric phenomena, such as thunderstorms and lightning.
3. Multipath: Occurs when multiple copies of the same signal reach the receiver through different paths, potentially causing measurement errors.
4. Interference: Occurs when an unwanted signal affects the desired radio signal.
5. Propagation Errors: Errors caused by variations in radio-wave propagation conditions compared with the assumptions used by the navigation system.

➢ **Why are these errors important in aviation?**

All these factors can affect the accuracy of bearing and distance measurements. Consequently, they may cause errors in the aircraft's estimated position or its deviation from the intended flight path.

System Design Considerations

When designing an aircraft navigation system, it is not enough for the system to simply receive radio signals. Several important performance criteria must be evaluated.

Criterion	Meaning
Accuracy	How close the measured value is to the true value.
Coverage	The geographical area in which the system can provide service.
Availability	The extent to which the service is available when needed.
Integrity	The ability of the system to warn users when an error exceeds acceptable limits.
Continuity	The ability of the system to provide uninterrupted service without unexpected interruptions.
Update Rate	How frequently navigation information is updated.
Reliability	The ability of the system to perform its intended function without failure.

Point Source Systems

The basic idea behind Point Source Systems is that the aircraft uses a single ground reference station for each basic measurement to determine a bearing, a distance, or both, depending on the system.

1. Direction-Finding System

A direction-finding system receives a radio signal and determines the direction in which the ground station is located relative to the aircraft.

❑ Two important angles are:

- Relative Bearing (RB): The angle between the aircraft's face and the direction of the ground station.
- True Bearing (TB): The direction of the ground station measured clockwise from true north.

If the aircraft's heading is known, then:

$$TB = (Heading + RB) \bmod 360^\circ$$

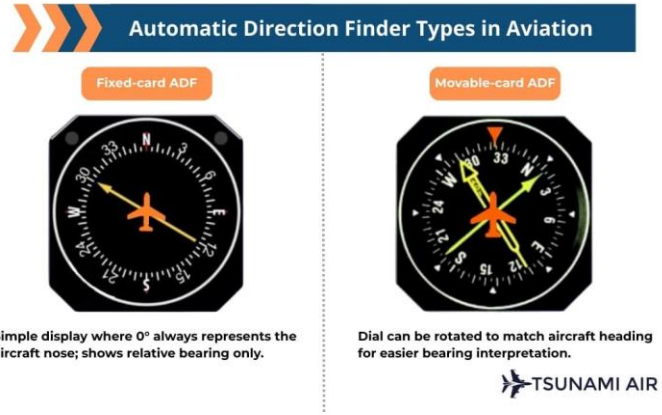
Example:

An aircraft has a heading of 070°, and the ground station is located 040° to the right of the aircraft.

Solution:

$$TB = 70^\circ + 40^\circ = 110^\circ$$

Therefore, the bearing of the station measured clockwise from true north is 110°.



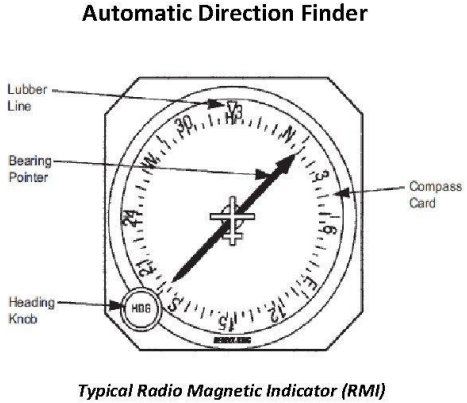
Note: This equation assumes that angles are measured clockwise and that a positive relative bearing indicates a direction to the right.

Point Source Systems

2.Non-Directional Beacon (NDB) and Automatic Direction Finder (ADF)

- **NDB — Ground Station**

A Non-Directional Beacon is a ground-based transmitter that broadcasts radio signals in multiple directions rather than transmitting a narrow beam toward a specific direction.



Typical Radio Magnetic Indicator (RMI)

- **ADF — Aircraft Equipment**

An Automatic Direction Finder receives the NDB signal and helps determine the direction of the station relative to the aircraft's face.

❑ Therefore, it is important not to confuse the two:

NDB: The ground-based transmitter.

ADF: The direction-finding equipment installed on the aircraft.

Limitations of the NDB/ADF system:

- It can be affected by atmospheric conditions and electrical interference.
- Radio propagation errors can affect bearing accuracy.
- The bearing indicated by the ADF does not necessarily represent the desired ground track.



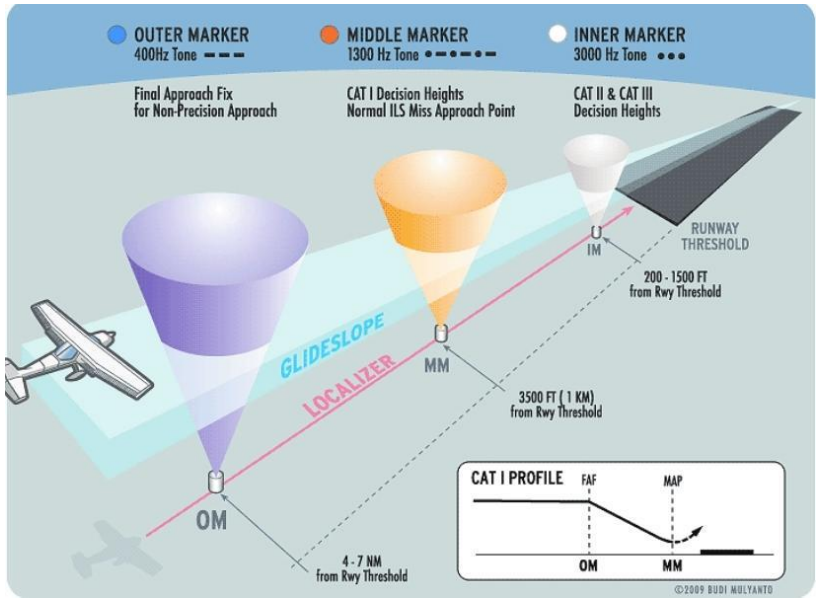
Point Source Systems

3. Marker Beacons

Marker beacons have traditionally been used in some aircraft approach and landing systems to identify specific points along the approach path.

When the aircraft passes over the signal coverage area, its receiver detects the signal and provides an indication to the pilot.

The primary function of a marker beacon is to identify a specific point along the approach path, rather than continuously measure distance as a DME system does.



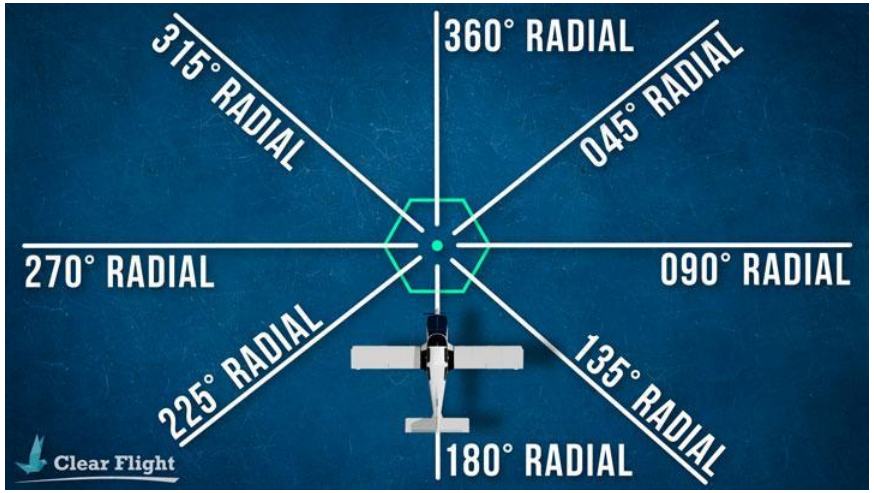
Point Source Systems

4. VHF Omnidirectional Range (VOR)

A VOR ground station transmits radio signals that allow the aircraft's receiver to determine its angular direction relative to the station.

The direction from the VOR station to the aircraft is called the **Radial**. It is measured clockwise from the magnetic north reference of the station.

For example, if an aircraft is on the 90° radial, it is located approximately east of the VOR station.





Point Source Systems

➢ How does VOR determine direction?

The system works by comparing the phases of two signals: a reference signal and a variable signal. The phase difference between these signals is related to the aircraft's angular position around the station. The receiver uses this phase difference to calculate the radial.

The essential concept is that the phase difference is converted into angular direction information.

➢ Advantages of VOR:

- Provides clear directional information.
- Helps aircraft follow defined airways and navigation routes.
- Can be combined with DME to provide both directional and distance information.

Limitations: VOR depends on radio coverage and line-of-sight propagation. Its accuracy can also be affected by signal reflections and interference.



Point Source Systems

5. Doppler VOR (DVOR)

DVOR is designed to reduce certain bearing errors caused by reflections from surrounding objects near the ground station.

It uses the Doppler principle to produce signals that enable the aircraft to determine its direction relative to the station.

➢ **The main educational difference is:**

- Conventional VOR: Generates the angularly varying signal using a conventional antenna and signal-generation arrangement.
- DVOR: Uses a different antenna arrangement and Doppler-based signal-generation technique to reduce certain site-related bearing errors.

✓ Both systems provide directional information, and neither is designed to measure distance directly.

✓ Distance information can be obtained by combining either system with DME.



Point Source Systems

6. Distance-Measuring Equipment (DME)

DME answers a different navigation question: **How far is the aircraft from the ground station?**

The aircraft transmits interrogation pulses, and the ground station responds with reply pulses. The aircraft's DME equipment measures the total elapsed time between transmitting the interrogation and receiving the reply. It then calculates the distance after compensating for the fixed delay inside the ground station.

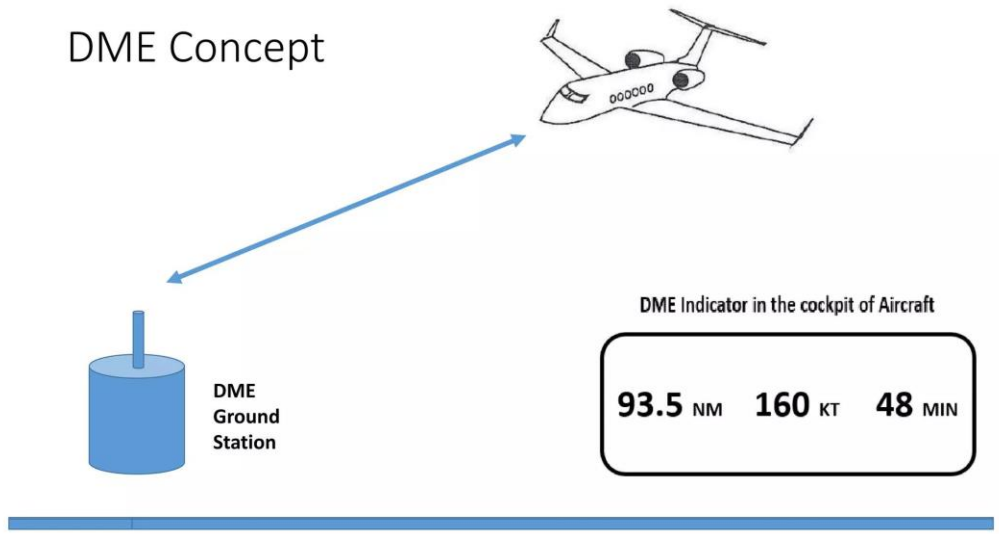
The basic equation is:

$$R = \frac{c}{2}(t_{\text{measured}} - t_{\text{delay}})$$

Where:

- R: Slant range between the aircraft and the ground station.
- c: Speed of light.
- t_{measured}: Total time between transmitting the interrogation and receiving the reply.
- t_{delay}: The specified delay introduced by the ground station.

✓ Important note: DME measures slant range, not horizontal distance directly. If an aircraft is flying at a considerable altitude and is horizontally close to the station, the difference between slant range and horizontal distance can be significant.



Point Source Systems

7. TACAN and VORTAC

TACAN (Tactical Air Navigation) is a navigation system that provides bearing and distance information. It is used primarily in military aviation.

VORTAC is a ground facility that combines two navigation systems:

- VOR: Provides bearing information.
- TACAN: Provides bearing and distance information to appropriately equipped aircraft.

Civilian aircraft equipped with a compatible DME receiver can obtain distance information from the distance-measuring service associated with the VORTAC facility, while using VOR for bearing information.



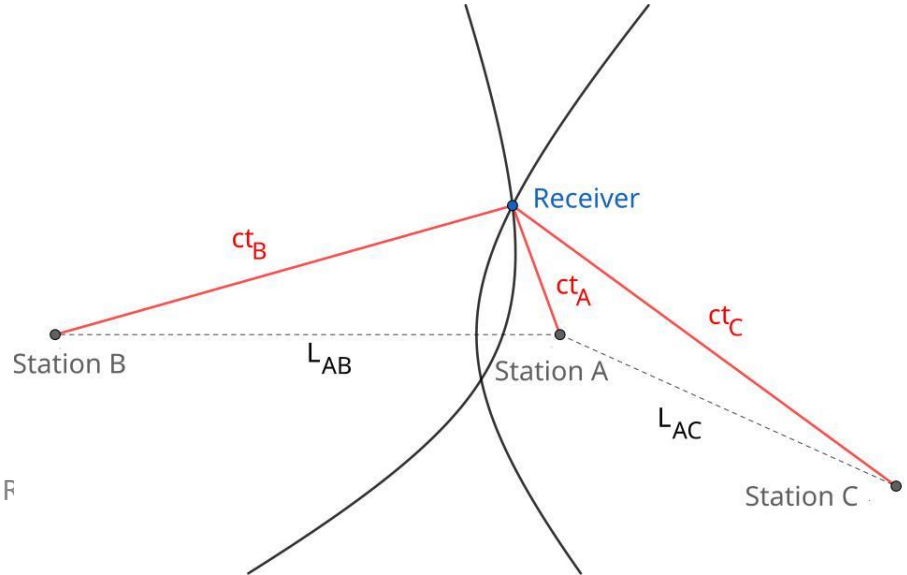
Hyperbolic Navigation Systems

Hyperbolic navigation is different from the systems discussed previously and is an important theoretical concept in radio navigation.

➢ What is Hyperbolic Navigation?

Instead of determining direction from a single station, a hyperbolic navigation system uses signals from two or more ground stations and compares their arrival times or phase differences.

If the difference between the distances from an aircraft to two stations is constant, the set of possible aircraft positions forms a hyperbola.



Hyperbolic Navigation Systems

Suppose there are two ground stations, A and B, and the aircraft is at point P. The distances are:

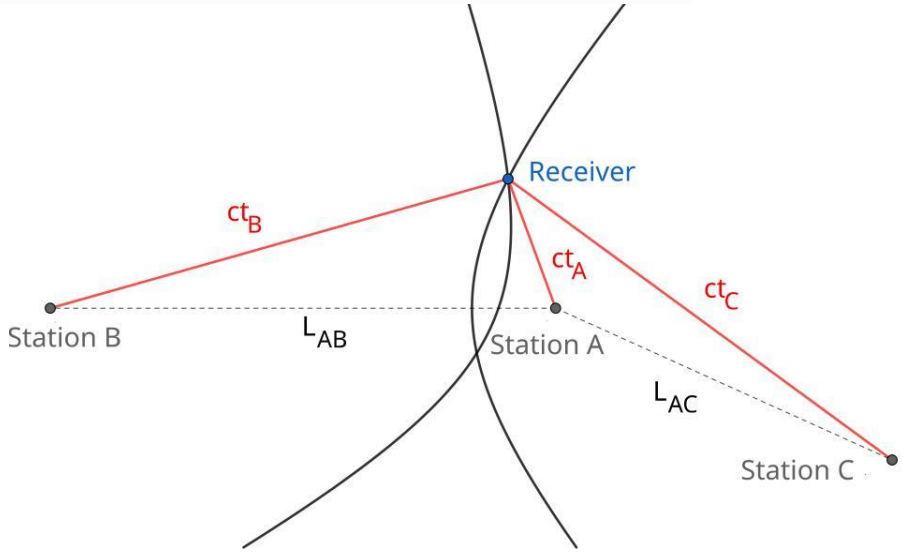
$$R_A = |PA|, \qquad R_B = |PB|$$

If the difference between these distances is known:

$$\boxed{\Delta R = R_A - R_B}$$

the aircraft lies on one branch of the hyperbola corresponding to that distance difference.

However, one distance-difference measurement does not uniquely determine the aircraft's position in two dimensions. An additional independent measurement is generally required to obtain another line of position and determine the aircraft's location from their intersection.



Hyperbolic Navigation Systems

Relationship Between Time Difference and Distance Difference

If two signals arrive at the aircraft with a time difference of Δt, the corresponding distance difference in the basic geometric model is:

$$\boxed{\Delta R = c\,\Delta t}$$

Notice the difference between this equation and the DME equation:

- **DME:** Measures the round-trip travel time, so the equation includes the factor $1/2$.
- **Hyperbolic navigation:** Uses the difference in signal arrival times, so the equation does not include the factor $1/2$.

Main Hyperbolic Navigation Systems

1. LORAN (Long Range Navigation)

LORAN uses synchronized ground-based transmitting stations, or stations with known timing relationships.
It determines lines of position by measuring differences in the arrival times of signals.

➢ **Operating Principle:**

- The aircraft receives signals from different ground stations.
- The receiver measures the time difference between the arrival of the signals.
- The time difference is converted into a difference in distance.
- The corresponding line of position is determined.
- Using another independent measurement, the aircraft's position can be determined from the intersection of the lines of position.

✓ Historically, one of the main advantages of LORAN-C was its ability to provide long-range navigation.
✓ Its limitations included dependence on ground-station infrastructure, precise timing, and the effects of signal propagation conditions on measurement accuracy.
✓ LORAN is now a historical navigation system in many regions and is not a modern, worldwide replacement for GNSS.

Main Hyperbolic Navigation Systems

2. Omega Navigation System

Omega was a global navigation system that used Very Low Frequency (VLF) signals.
It estimated position by comparing the phases of signals received from multiple ground stations.
Its operating principle differs from that of LORAN-C, which is based on measuring differences in the arrival times of pulses.
Omega primarily relied on phase measurements.

- ✓ The Omega system was shut down in 1997. Therefore, it is studied today as an important historical system in understanding the development of radio navigation.

3. Decca Navigation System

Decca relied on comparing the phases of signals transmitted by ground stations to determine curved lines of position.
Historically, it was used extensively in **maritime navigation**.
The main concept should understand is that a phase difference, like a time difference, can be converted into information that identifies a set of possible positions.
Additional measurements are then used to determine the actual position.

Main Hyperbolic Navigation Systems

4. Chayka Navigation System

Chayka is a historical ground-based navigation system that operates on a principle similar to that of the LORAN family.
It was used in the former Soviet Union and Russia.
The system relies on signal-timing measurements between ground stations to establish lines of position.
Its fundamental mathematical principle is to convert the measurements into differences in distance and then calculate the receiver's position.

Radio-wave power and the basic link equation

- The received signal becomes weaker as the distance between transmitter and receiver increases.
- A basic free-space relationship is related to the **Friis transmission equation**:

$$\boxed{P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2}$$

where:
P_t= transmitted power
P_r= received power
G_t= transmitting antenna gain
G_r= receiving antenna gain
λ= wavelength
R= distance between transmitter and receiver
and

$$\boxed{\lambda = \frac{c}{f}}$$

Therefore, frequency is directly related to wavelength.

$\boxed{v = c}$ in free space.

Therefore: $\boxed{v \approx 3 \times 10^8 \text{ m/s}}$

Radio-wave power and the basic link equation

This is extremely important for navigation systems that determine **distance from signal travel time**.

For example: $\boxed{R = ct}$ for one-way propagation.

For a round trip: $\boxed{R = \frac{ct}{2}}$

This is exactly the principle that you seen in **DME**.

At distance R, the area of the sphere is: $\boxed{A_{\text{sphere}} = 4\pi R^2}$

Therefore, as the signal travels farther away, the same transmitted energy is spread over a larger area.

If the transmitter radiates power P_t isotropically, the power density at distance R is:

$$\boxed{S = \frac{P_t}{4\pi R^2}}$$

where:

S= power density (W/m^2)

P_t= transmitted power

R= distance

Radio-wave power and the basic link equation

Now imagine that the receiving antenna has an effective area A_r.

The received power is: $P_r = SA_r$

Therefore:

$$P_r = \frac{P_t}{4\pi R^2} A_r$$

So:

$$\boxed{\frac{P_r}{P_t} = \frac{A_r}{4\pi R^2}}$$

Radio-wave power and the basic link equation

➢ **What happens when distance increases?**

From the equation: $P_r \propto \frac{1}{R^2}$

approximately, under ideal free-space conditions. So:

Distance ↑ >> Received power ↓ >> Signal-to-noise ratio ↓ >> Navigation accuracy/reliability can deteriorate

This is why **coverage range** is an important design parameter.

The receiving antenna doesn't simply have a physical size.

uses the concept of **effective area**, or **effective aperture**: $\boxed{A_e}$

It represents the effective area through which the antenna can collect electromagnetic energy.

Thus: **Pr = S Ae**

➢ An antenna can concentrate radiation in particular directions.

Instead of spreading energy equally in all directions, a directional antenna can concentrate more energy in the desired direction.

Conceptually:

- **Isotropic antenna**
- **Directional antenna**

Radio-wave power and the basic link equation

Directivity and Efficiency $\boxed{G = D\eta}$

G= antenna gain
D= directivity
η= antenna efficiency
So gain depends on:

- how directional the antenna is
- how efficiently it converts input power into radiation.

If $\eta = 1$ then $G = D$ for an ideal lossless antenna.

maximum effective aperture to directivity: $\boxed{A_e = \frac{D\lambda^2}{4\pi}}$

For a practical antenna, since: $G = D\eta$
the commonly used relationship is:

$$\boxed{A_e = \frac{G\lambda^2}{4\pi}}$$

This equation is extremely important because it connects:

Antenna gain ↔ effective area ↔ wavelength

Radio-wave power and the basic link equation

in terms of the effective areas of both antennas.
The result is:

$$\boxed{\frac{P_r}{P_t} = \frac{A_t A_r}{\lambda^2 R^2}}$$

where:
A_t= effective area of transmitting antenna
A_r= effective area of receiving antenna
R= distance
λ= wavelength

Effective Radiated Power

$$\boxed{ERP = P_t G_t}$$

where:
P_t= transmitter power
G_t= transmitter antenna gain
This is called the **effective radiated power** in the direction of maximum gain.

Radio-wave power and the basic link equation

➢ Receiver Sensitivity

Now moves from the transmitter side to the receiver side.

The **minimum signal power that a receiver can detect** is called: Receiver sensitivity

A receiver might have enormous amplification, but amplification cannot solve everything.

Why? Because the receiver itself contains **noise**.

➢ Two Types of Noise

The book divides noise into

❑ **External noise** : N_{external}

Coming from outside the receiver:

- unwanted transmitters,
- electrical machinery,
- atmospheric noise,
- cosmic noise,

❑ **Internal noise**

Generated within the receiver electronics.

The fundamental lower limit is related to **thermal noise**.

Radio-wave power and the basic link equation

- **Thermal Noise**

$$N_p = kT\Delta f$$

N_p= noise power in watts
k= Boltzmann's constant
T= absolute temperature in kelvin
Δf= bandwidth in Hz
Boltzmann's constant is: k=1.38×10−23J/K

the receiver noise figure in terms of the practical output noise compared with the ideal thermal-noise output:

$$NF = \frac{N_o}{kT_o\Delta f\ G_a}$$

N_o= noise power out of practical receiver
k= Boltzmann constant
T_o= standard reference temperature
Δf= bandwidth
G_a= available receiver gain

The ideal receiver output noise would be:

$$N_{o,\text{ideal}} = kT_o\Delta f G_a$$

$$NF = \frac{\text{actual output noise}}{\text{ideal output noise}}$$

Radio-wave power and the basic link equation

➢ **Noise Figure in dB** $\boxed{NF_{dB} = 10\log_{10}(NF)}$

For example, if: $NF = 2$

$$NF_{dB} = 10\log_{10}(2)$$

$$\boxed{NF_{dB} \approx 3.01\ dB}$$

A lower noise figure is desirable because it means the receiver adds less noise.

➢ **Spectrum Spreading**

This is an important idea.

Suppose the actual navigation information requires only: $B_{\text{information}}$

but the system intentionally spreads the signal over:

$$B_{\text{system}} > B_{\text{information}}$$

This may seem inefficient, but it can provide advantages such as:

Improved resistance to interference

and:

Improved SNR/measurement performance

This concept becomes especially important later when you study **GPS**.

Radio-wave power and the basic link equation

➢ The Link Budget

Given transmitter power, antenna gains, receiver noise, bandwidth, required S/N, and propagation loss, how much transmitter power is required?

The generalized link-budget expression is given in dB form as:

$$\frac{P_t}{P_n} = \left(\frac{S}{N}\right)_{req} + L_p + NF - G_t - G_r - F_n$$

with the terms interpreted in dB.

P_t= transmitter power

P_n= receiver noise power

$(S/N)_{req}$= required signal-to-noise ratio

NF= receiver noise figure

F_n= noise-improvement factor associated with modulation/bandwidth spreading

G_t= transmitter antenna gain

G_r= receiver antenna gain

L_p= propagation path attenuation loss

Radio-wave power and the basic link equation

➢ Propagation Loss

For free-space propagation, the familiar loss relationship is related to:

$$\boxed{L_p \propto \left(\frac{4\pi R}{\lambda}\right)^2}$$

In dB:

$$\boxed{L_p(dB) = 20\log_{10}\left(\frac{4\pi R}{\lambda}\right)}$$

Using:

$$\lambda = \frac{c}{f}$$

gives:

$$\boxed{L_p(dB) = 20\log_{10}\left(\frac{4\pi R f}{c}\right)}$$

Radio-wave power and the basic link equation

Example :

Suppose a system has:
propagation loss = 120 dB
receiver noise figure = 5 dB
required S/N = 10 dB
transmitter antenna gain = 15 dB
receiver antenna gain = 5 dB
Then the approximate link requirement becomes:

$$\text{Required link} = 10 + 120 + 5 - 15 - 5$$

$$\boxed{= 115\ dB}$$

The advantage of dB is that multiplication/division in the original power equations becomes addition/subtraction.

Propagation and Noise Characteristics

- The radio signal doesn't simply travel through empty space.

Its propagation depends on:

- frequency,
- Earth's surface,
- atmosphere,
- ionosphere,
- distance,
- terrain,
- precipitation,
- atmospheric absorption,
- scattering,
- noise.

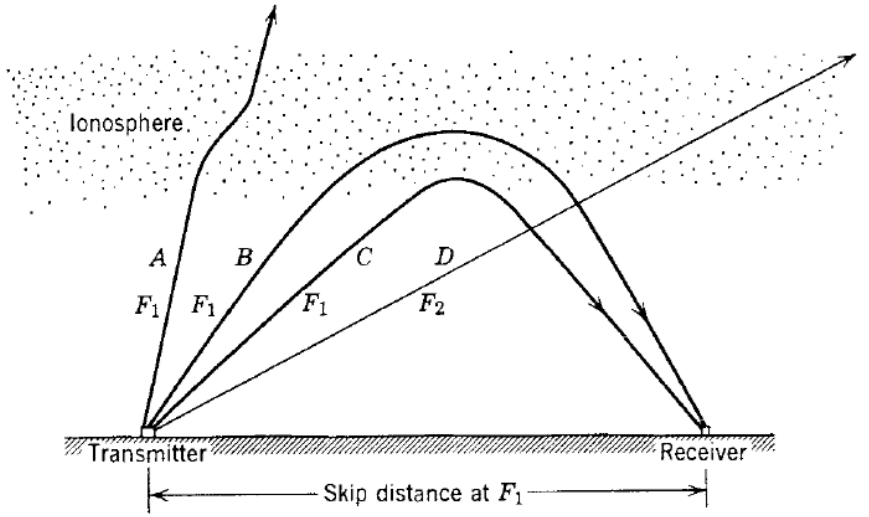


Figure 4.2 Effects of ionosphere on radio waves.

Propagation and Noise Characteristics

➢ Ground Waves

At relatively low frequencies, radio waves can propagate close to the Earth's surface.
This is called:
Ground-wave propagation
The signal follows the Earth's surface to some extent.
The ground-wave attenuation depends strongly on:

- frequency,
- electrical conductivity of the ground,
- terrain,
- distance.

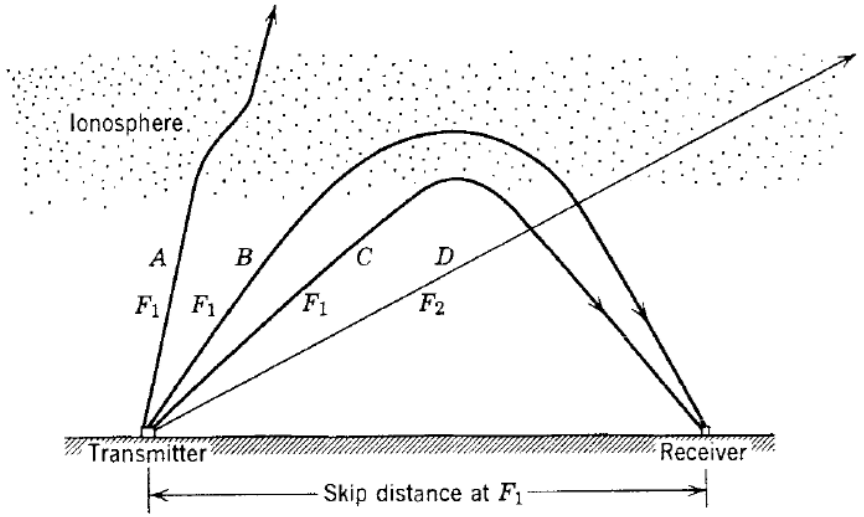


Figure 4.2 Effects of ionosphere on radio waves.

Propagation and Noise Characteristics

- **Why seawater is important ?**

Seawater has much higher electrical conductivity than ordinary land.
Therefore, ground-wave propagation can be much better over seawater than over poor-conductivity land.
This becomes relevant for long-range terrestrial radio systems.

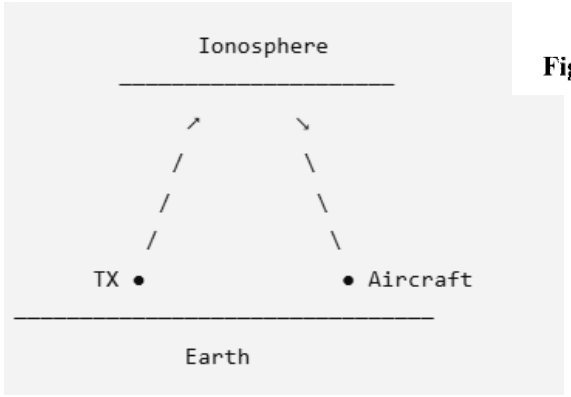
- **Sky Waves**

At lower frequencies, radio signals can also interact with the ionosphere.

The wave can be refracted/returned by ionospheric layers.
This produces: Sky–wave propagation
The problem is that ionospheric conditions vary.
They can depend on:

- time of day,
- season,
- geographic location,
- solar activity,
- frequency.

Therefore, sky-wave propagation can introduce uncertainty into navigation measurements.



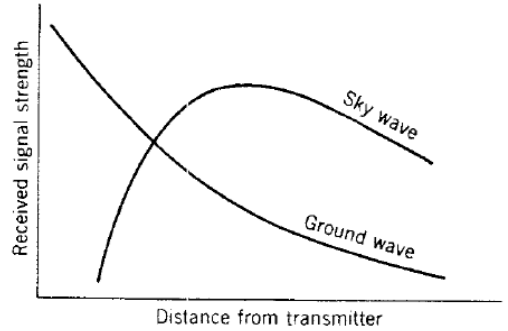


Figure 4.3 Ground-wave and sky-wave attenuation of radio signals.

Propagation and Noise Characteristics

➢ Skip Distance

This is related to the minimum ground distance at which an ionospherically returned signal can be received.
There may be a region near the transmitter where the sky wave is not returned to the surface.
This is the **skip zone**.

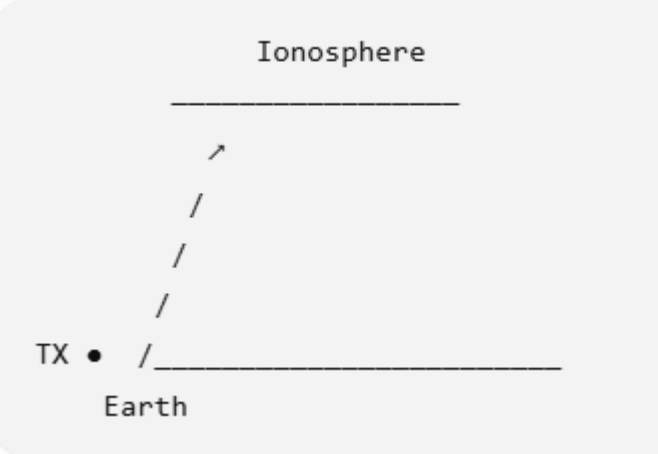
➢ Noise

A navigation receiver does not receive only the desired signal.

It receives: Signal + Noise + Interference

❑ Noise can come from:

- **Atmospheric noise :** Lightning and atmospheric electrical activity.
- **Man-made noise :** Electrical equipment, power systems, vehicles, etc.
- **Receiver noise :** Noise generated internally by the electronics.



Propagation and Noise Characteristics

➢ **Signal-to-Noise Ratio**

One of the fundamental concepts is:

$$SNR = \frac{P_S}{P_N}$$

where:

P_S= signal power

P_N= noise power

In dB:

$$SNR_{dB} = 10\log_{10}\left(\frac{P_S}{P_N}\right)$$

Higher SNR generally makes it easier for the receiver to determine the navigation measurement accurately.

Propagation and Noise Characteristics

- **Why Frequency Matters**

Frequency → Propagation characteristics → System design

Different frequencies behave differently.

Frequency region	Important behavior
Low frequency	Ground-wave propagation, ionospheric effects
VHF	Relatively predictable, line-of-sight
Higher frequencies	More atmospheric absorption/scattering

At sufficiently high frequencies, atmospheric effects such as precipitation and absorption become increasingly important.

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Thank You!

See You next Lectures!!
Any Question?

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